

Strategic Risk in Repeated Games with Imperfect Monitoring*

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Abstract

In this paper, I study the trade-off between efficiency and strategic risk in a repeated Prisoner's Dilemma with imperfect public monitoring. Because bad signals can arise on the equilibrium path, sustaining a high value of cooperation requires players to adopt more lenient and forgiving strategies, which in turn makes them more vulnerable to opportunistic behavior. For a broad class of one-dimensional public signals, I show that no single strategy both maximizes the value of cooperation and minimizes exposure to strategic risk. This contrasts with standard results under perfect monitoring, where strategic risk can determine whether players cooperate at all, but not which cooperative strategy they choose. Using experimental data from Aoyagi and Fréchette (2009), I document suggestive patterns consistent with subjects choosing strategies to maximize the value of cooperation rather than limit their exposure to strategic risk. **Keywords:**

Repeated Games; Imperfect Public Monitoring; Strategic Risk.

JEL: C72; C73; C91; D82

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I INTRODUCTION

In an uncertain world, where actions are often observed with noise, understanding the tension between opportunistic and cooperative behavior is at the core of economics, evolutionary biology, and any discipline concerned with social interactions.

In the context of a repeated Prisoner's Dilemma, both under perfect and imperfect monitoring, the folk theorems predict that, if players are patient enough, many cooperative equilibrium outcomes can arise (e.g. Fudenberg et al., 1994). However, a stark difference emerges between the two monitoring structures. Under perfect monitoring, provided that the incentives to cooperate are strong enough, the way off-path punishments are structured does not affect the payoffs attained on the cooperative path. Conversely, when monitoring is imperfect, bad realizations of the signal can happen after a sequence of mutual cooperations on the equilibrium path. Therefore, for the repeated interaction to be valuable, not only do players have to establish cooperation as fast as possible, but they also have to be lenient (not punish for sufficiently good signals) and forgiving (give cooperation a second chance after a period of punishment and a good realization of the signal) in order to avoid inefficient breaches.¹ However, given the uncertainty about the strategies adopted by the opponent, increasing the levels of leniency and forgiveness exposes the players to opportunistic behaviors and to defectors.

In this paper, I show that under imperfect monitoring no single strategy simultaneously maximizes the value of cooperation and minimizes strategic risk. This contrasts with perfect monitoring, where the strategy with the harshest punishment, Grim Trigger, achieves both the highest value of cooperation and the lowest strategic risk (Blonski & Spagnolo, 2015). This difference generates predictions that are unavailable under perfect monitoring. There, strategic risk operates only on the extensive margin, determining whether players cooperate or instead coordinate on the uniquely safe always-defect equilibrium.² Under imperfect monitoring, by contrast, strategic risk also shapes the intensive margin, determining which cooperative strategy players select conditional on cooperating.

Formally, I consider a repeated symmetric Prisoner's Dilemma with two players and discount factor δ . In each period, the action profile determines the continuous distribution of a one-dimensional public signal observed by both players. In line with

¹The experimental findings in Fudenberg et al. (2012) show that, under imperfect public monitoring, players tend to adopt more forgiving and lenient strategies.

²Always defect is the uniquely safe equilibrium strategy because its payoff along the symmetric equilibrium path is a lower bound on the payoff it obtains when the opponent unilaterally deviates from always defect.

previous research (Aoyagi & Fréchette, 2009; Green & Porter, 1984), I assume that higher realizations of the public signal are more likely when the opponent chooses the cooperative action. I focus on symmetric public perfect equilibria (henceforth *S-PPE*), in which strategies condition only on public histories and constitute a Nash equilibrium after every such history. The main measure of strategic risk is pairwise p -dominance relative to AllD, formalized for a special case in Blonski and Spagnolo (2015). A strategy is pairwise p -dominant relative to AllD if it is played in a symmetric equilibrium and is a best response when the opponent plays that strategy with probability p and AllD with probability $1 - p$.

The first contribution is a frontier result for the full class of *S-PPE* strategies. If cooperation can be sustained in a pairwise p -dominant equilibrium, then, for every belief $p < 1$, the maximal payoff attainable in an unconstrained equilibrium is strictly higher than the maximal payoff attainable among pairwise p -dominant equilibria. Although this may appear intuitive, since pairwise p -dominance imposes an additional constraint on the equilibrium payoff set, the result contrasts sharply with the case of perfect monitoring, where the constraint is not binding. Under perfect monitoring, once a pairwise p -dominant equilibrium exists, the strategy that attains the maximal equilibrium payoff remains the same pairwise p -dominant strategy, namely grim trigger, and no payoff loss is incurred.

While a general characterization of the payoff-maximizing strategy subject to pairwise p -dominance is intractable, I test the theoretical framework by restricting attention to two classes of strategies for which I derive the exact equilibrium and pairwise p -dominant payoff frontiers. In the threshold-Grim class, cooperation is sustained as long as the public signal remains above a cutoff; once the signal falls below that cutoff, play switches permanently to punishment. In the two-state public-forgiving class, punishment is instead reversible: the strategy enters the punishment state after a sufficiently low signal in the cooperation state and returns to cooperation after a sufficiently high signal in the punishment state.

Exploiting these classes of strategies, I provide a structural reanalysis of the experiment in Aoyagi and Fréchette (2009). I estimate, separately for each noise treatment, a finite-mixture model that assigns subjects to one of three public strategies: AllD, public Grim, and public forgiving. For the latter two types, the model also estimates the subject-specific signal thresholds that govern transitions into, and in the forgiving case out of, punishment. I then use these estimated strategies to compute the counterfactual values that subjects would obtain in symmetric self-play.

The main finding is that many of the estimated strategies generate simulated self-play payoffs close to the maximal payoff attainable in an *S-PPE* and, in some cases, even above

it. These strategies generate highly efficient outcomes because they keep punishments short and allow cooperation to resume quickly after bad signals. While p -dominance is a good predictor of whether subjects adopt a cooperative strategy at all under perfect monitoring (e.g., Blonski et al., 2011; Dal Bó and Fréchette, 2018), it appears to be a poor predictor of which cooperative strategy they choose under imperfect monitoring. Instead, strategy choice is more closely related to the maximal attainable equilibrium value.

RELATED LITERATURE

The closest related paper is Aoyagi and Fréchette (2009), in which the authors characterize the set of maximal payoffs attained in an S -PPE for the class of repeated Prisoner’s Dilemmas with the same imperfect monitoring structure as in this paper. Aoyagi and Fréchette estimate public threshold strategies and show that the value of cooperation declines as monitoring becomes noisier, in line with theoretical predictions. Moreover, subjects seem to adopt lenient and forgiving strategies, sustaining a high cooperation value. Fudenberg et al. (2012) find similar results using a two-dimensional, discrete signal structure.³ This paper complements those findings by formalizing the trade-off between the value of cooperation and strategic risk and deriving testable predictions for how lenient and forgiving agents can be under different equilibrium selection criteria.

In the class of repeated Prisoner’s Dilemmas with perfect monitoring, Blonski et al. (2011) provide an axiomatic theory of equilibrium selection that predicts, at a coarse level, whether players adopt a cooperative strategy or the always-defect equilibrium. Specifically, they establish a threshold for the discount factor (δ^*) such that players are predicted to choose the cooperative equilibrium when $\delta > \delta^*$. This threshold captures the intuitive idea, absent from standard equilibrium analysis, that cooperation becomes riskier when the “sucker’s payoff” from cooperating against defection decreases. In a related experiment, they show that (i) when $\bar{\delta}$, the threshold for existence of a subgame-perfect equilibrium (SPE), and δ^* move in opposite directions, cooperation changes as predicted by δ^* ; and (ii) cooperation remains low when $\bar{\delta} < \delta < \delta^*$.

In related work, Blonski and Spagnolo (2015) show that the critical discount factor of Grim Trigger coincides with the axiomatically founded δ^* in Blonski et al. (2011). The intuition is that Grim attains the maximal cooperation value when matched with itself while limiting the loss against always defect. With imperfect monitoring, however,

³In particular, after choosing the intended action, there is a probability ε that each player’s choice is changed to the other action. The implemented choices determine payoffs, and both players are informed of those choices.

Bigoni et al. (2024) show that δ^* need not be a lower bound for cooperative equilibria to be pairwise p -dominant at $p = 1/2$. Grim's limited loss against always defect need not compensate for the lower value of cooperation, which enlarges its basin of attraction relative to other equilibria. While no lower bound for the discount factor is currently known in settings with discrete noise, this paper advances the literature by formalizing the trade-off between strategic risk and the value of cooperation under imperfect monitoring. The paper therefore analyzes equilibrium selection at a finer level by developing a theory of which strategies can support cooperation in a pairwise p -dominant equilibrium.

II THEORETICAL FRAMEWORK

I consider an infinitely repeated Prisoner's Dilemma with imperfect public monitoring. At each stage, player i chooses an action a_i from the set $A_i = \{c, d\}$, where c is the cooperative action and d is the noncooperative one. The set of action profiles $a = (a_i, a_j)$ is $A = A_i \times A_j$. In each period, both players observe a random public signal $z \in \mathbb{R}$, conditional on the implemented action profile a according to a density function $p(z \mid a)$. The deterministic case corresponds to perfect public monitoring when the signal map reveals the payoff-relevant action profile.

Assumption 1 (Public-signal regularity). The densities $p(\cdot \mid a)$ have a common interval support $Z \subseteq \mathbb{R}$, are continuous, and are strictly positive on the interior of Z . Their tail probabilities are continuous and converge to one and zero at the lower and upper endpoints of Z , respectively.

Let $u^{a_i, a_j} := u(a_i, a_j)$ denote player i 's stage payoff when player i chooses a_i and player j chooses a_j . Expected stage payoffs satisfy

$$u^{dc} > u^{cc} > u^{dd} > u^{cd}. \quad (1)$$

The monitoring technology and the stage game are symmetric. A public history at date t is $h^{P,t} = (z^0, \dots, z^{t-1}) \in Z^t$, with $Z^0 = \{\emptyset\}$. A pure public strategy σ_i is a Borel-measurable map from public histories to $\{c, d\}$. Payoffs are normalized:

$$\pi_i(\sigma_i, \sigma_j) = (1 - \delta) \mathbb{E}_{\sigma_i, \sigma_j} \left[\sum_{t=0}^{\infty} \delta^t u_i(a_i^t, a_j^t) \right].$$

I use the equilibrium notion of symmetric public perfect equilibrium (S -PPE): both players use the same public strategy after every public history, and the resulting strategy profile constitutes a Nash equilibrium after every such history.

In line with previous research (Aoyagi and Fréchet, 2009; Green and Porter, 1984), I make two further assumptions. The first implies that the action profiles $a = (c, d)$ and $a = (d, c)$ induce the same density function for the public signal. The second postulates that a higher realization of the public signal corresponds to an increased likelihood of the opponent choosing the cooperative action.

Assumption 2 (Symmetric monitoring). For every $z \in Z$,

$$p(z \mid cd) = p(z \mid dc).$$

Assumption 3 (Strict monotone likelihood ratios). For every $z' > z$ in Z ,

$$\frac{p(z' \mid cc)}{p(z \mid cc)} > \frac{p(z' \mid cd)}{p(z \mid cd)} = \frac{p(z' \mid dc)}{p(z \mid dc)} \geq \frac{p(z' \mid dd)}{p(z \mid dd)}.$$

Let the set of equilibrium strategies for a given discount factor δ be

$$\mathcal{S}_\delta = \{\sigma : (\sigma, \sigma) \text{ is a pure } S\text{-PPE}\}$$

and let $v(\sigma) = \pi_i(\sigma, \sigma)$. The strategy that defects after every history is denoted by AllD, which stands for “*always defect*”. It belongs to $\mathcal{S}_\delta$ and yields $v(\text{AllD}) = u^{dd}$. For public strategies σ and τ , write $V_{\sigma, \tau} = \pi_i(\sigma, \tau)$.

Definition 1. Fix $p \in (0, 1]$. An equilibrium (σ, σ) is *pairwise p -dominant relative to AllD* if $\sigma \in \mathcal{S}_\delta$ satisfies

$$p v(\sigma) + (1 - p) V_{\sigma, \text{AllD}} \geq p V_{\text{AllD}, \sigma} + (1 - p) u^{dd}. \quad (2)$$

The intuition is that a cooperative strategy can be sustained in a pairwise p -dominant equilibrium when the expected gains from coordinating on that strategy outweigh the expected losses from facing an opponent who plays AllD. While the standard notion of p -dominance (Morris et al., 1995) requires the candidate strategy to be a best response to any conjecture assigning probability at least p to the equilibrium strategy, the pairwise p -dominance criterion used here restricts attention to the reduced strategy set $\{\sigma, \text{AllD}\}$.⁴ In this reduced two-strategy game, the minimal p for which σ is pairwise p -dominant corresponds to the size of the basin of attraction of AllD, commonly referred to as

⁴This pairwise restriction parallels the approach of Blonski and Spagnolo (2015), who extend risk-based equilibrium selection to repeated games by comparing a cooperative equilibrium with the uniquely safe AllD equilibrium. Standard p -dominance implies pairwise p -dominance, but the converse need not hold.

SizeBAD in the experimental repeated-games literature.

Let the set of strategies that are sustainable in a pairwise p -dominant equilibrium with discount factor δ be

$$\mathcal{S}_\delta^p = \{\sigma \in \mathcal{S}_\delta : \sigma \text{ satisfies (2)}\}.$$

The two payoff frontiers are

$$\bar{v}^{\text{PPE}} = \sup_{\sigma \in \mathcal{S}_\delta} v(\sigma), \quad \bar{v}^p = \sup_{\sigma \in \mathcal{S}_\delta^p} v(\sigma).$$

Under perfect monitoring, when $\bar{v}^p > u^{dd}$, the two frontiers coincide. In that case, the grim trigger strategy both maximizes the symmetric PPE payoff and satisfies the pairwise p -dominance restriction: after any observed deviation, grim trigger moves permanently to punishment, so the threat that supports cooperation also makes the cooperative path robust to the belief perturbations captured by pairwise p -dominance. Thus the additional pairwise p -dominance requirement does not reduce the maximal attainable payoff.

In the next section I show that under imperfect public monitoring, this equivalence breaks down. Bad public signals are only noisy evidence of deviations, so stronger deterrence requires exposing players to more punishment following low signals, including signals that may arise after cooperative play. The strategy that maximizes the PPE payoff therefore does not provide enough deterrence to satisfy pairwise p -dominance, and imposing pairwise p -dominance strictly lowers the attainable payoffs.

III THEORETICAL RESULTS

For any public history $h^{P,t}$, let $q_\sigma(h^{P,t}) = 1$ if σ prescribes c and zero otherwise. The normalized discounted exposure of σ against AllD is

$$x(\sigma) = (1 - \delta) \mathbb{E}_{\sigma, \text{AllD}} \left[\sum_{t=0}^{\infty} \delta^t q_\sigma(h^{P,t}) \right] \in [0, 1]. \quad (3)$$

This measures the discounted frequency with which the candidate strategy cooperates along the histories generated by an opponent who always defects.

Lemma 1 (Payoff–exposure representation). *Under Assumption 2, $\sigma \in \mathcal{S}_\delta$ is pairwise p -dominant if and only if*

$$v(\sigma) - u^{dd} \geq \chi(p)x(\sigma), \quad \chi(p) = u^{dc} - u^{dd} + \frac{1-p}{p}(u^{dd} - u^{cd}). \quad (4)$$

Lemma 1 reduces the pairwise p -dominance requirement to a simple linear restriction in payoff–exposure space. The first coordinate, $v(\sigma)$, is the continuation payoff generated by the strategy, while the second, $x(\sigma)$, measures the strategy’s average exposure to unilateral defection incentives. The coefficient $\chi(p)$ depends only on the stage-game payoffs and the belief p . This representation separates the feasible consequences of public strategies from the incentive discipline imposed by pairwise p -dominance. Define the feasible payoff–exposure set as

$$\mathcal{E}_\delta = \{(v(\sigma), x(\sigma)) : \sigma \in \mathcal{S}_\delta\}.$$

To ensure that the resulting problem is well behaved, I impose the following compactness assumption.

Assumption 4 (Compact payoff–exposure set). $\mathcal{E}_\delta$ is compact in $\mathbb{R}^2$.

Under this assumption, the unconstrained payoff frontier and the pairwise p -dominant frontier are both attained by feasible payoff–exposure pairs.

Theorem 1. *Maintain Assumptions 1 to 4. If $\bar{v}^{\text{PPE}} > u^{dd}$, then, for every $p \in (0, 1)$, $\bar{v}^p < \bar{v}^{\text{PPE}}$.*

The theorem shows that, under imperfect public monitoring, the payoff-maximizing S -PPE is not robust to strategic uncertainty. A player who seeks the highest S -PPE value must rely on a strategy that is too fragile to strategic uncertainty, whereas a player who insists on pairwise p -dominance must accept a lower attainable payoff. This contrasts with perfect monitoring, where Grim Trigger, whenever it is p -dominant, also attains the maximal cooperative payoff.

Importantly, the result does not characterize the full set of pairwise p -dominant equilibria, nor does it imply that the trade-off exists within all classes of cooperative equilibria. To make this trade-off operational, the analysis below restricts attention to two subclasses of public strategies for which the relevant payoff–risk frontiers can be characterized explicitly and the trade-off can be established. These classes are the imperfect-monitoring counterparts of two standard symmetric strategies studied under perfect monitoring: Grim Trigger and Tit-for-Tat (Dal Bó & Fréchette, 2018).

III(i) THRESHOLD-GRIM STRATEGIES

The threshold-Grim strategy cooperates initially and continues to cooperate as long as all previous public signals are at least k . Following any signal below k , it switches

permanently to AllD. Fix a threshold-Grim strategy σ with cutoff k . Its symmetric self-play value is

$$v(\sigma) = u^{dd} + \frac{(1 - \delta)(u^{cc} - u^{dd})}{1 - \delta\mathbb{P}(z \geq k \mid cc)}. \quad (5)$$

Aoyagi and Fréchette (2009) show that a threshold-Grim strategy σ with cutoff k is a PPE if and only if

$$\Lambda_\delta(k) := \frac{1 - \delta\mathbb{P}(z \geq k \mid dc)}{1 - \delta\mathbb{P}(z \geq k \mid cc)} \geq \frac{u^{dc} - u^{dd}}{u^{cc} - u^{dd}}.$$

They also show that, under Assumptions 1–3, the maximal S -PPE payoff $\bar{v}^{\text{PPE}}$ can be attained by a threshold-Grim strategy.

Proposition 1. *Maintain Assumptions 1 and 2. The threshold-Grim strategy σ with cutoff k is pairwise p -dominant if and only if*

$$\Lambda_\delta(k) \geq \frac{\chi(p)}{u^{cc} - u^{dd}}.$$

The sets of thresholds that satisfy, respectively, the S -PPE and pairwise p -dominance conditions are

$$K(\delta) = \left\{ k : \Lambda_\delta(k) \geq \frac{u^{dc} - u^{dd}}{u^{cc} - u^{dd}} \right\}, \quad K^p(\delta) = \left\{ k : \Lambda_\delta(k) \geq \frac{\chi(p)}{u^{cc} - u^{dd}} \right\}.$$

Under the maintained assumptions Assumptions 1 to 3, whenever $K(\delta)$ and $K^p(\delta)$ are nonempty, both sets are compact intervals. Let

$$k^*(\delta) = \min K(\delta), \quad k^p(\delta) = \min K^p(\delta).$$

These cutoffs attain the equilibrium and pairwise p -dominant value frontiers within the threshold-Grim class.

Theorem 2. *Under Assumptions 1 to 3, if $K(\delta)$ and $K^p(\delta)$ are nonempty and $p < 1$, then*

$$k^p(\delta) > k^*(\delta).$$

Since the value in equation (5) is decreasing in k , it follows that the maximal value attainable in an S -PPE within the threshold-Grim class is strictly higher than the maximal value attainable under pairwise p -dominance whenever $p < 1$.

This restricted-class characterization shows that the strict frontier separation established for the full class of symmetric public strategies also arises within the threshold-Grim

class. Whenever both sets are nonempty and $p < 1$, imposing pairwise p -dominance strictly raises the value-maximizing cutoff and lowers the maximal attainable payoff.

III(ii) TWO-STATE PUBLIC FORGIVING STRATEGIES

The second class we consider consists of public strategies with two states: a cooperation state C and a punishment state P . In C , both players choose c ; after $z \leq k_l$, the automaton enters P . In P , both choose d ; after $z > k_h$, it returns to C . Let $v_C(k_l, k_h)$ and $v_P(k_l, k_h)$ be the symmetric values from the two states. They satisfy

$$v_C(k_l, k_h) = (1 - \delta)u^{cc} + \delta\mathbb{P}(z > k_l \mid cc)v_C(k_l, k_h) + \delta\mathbb{P}(z \leq k_l \mid cc)v_P(k_l, k_h), \quad (6)$$

$$v_P(k_l, k_h) = (1 - \delta)u^{dd} + \delta\mathbb{P}(z > k_h \mid dd)v_C(k_l, k_h) + \delta\mathbb{P}(z \leq k_h \mid dd)v_P(k_l, k_h). \quad (7)$$

The strategy is an S -PPE if and only if both

$$(1 - \delta)(u^{cc} - u^{dc}) + \delta[\mathbb{P}(z > k_l \mid cc) - \mathbb{P}(z > k_l \mid dc)][v_C(k_l, k_h) - v_P(k_l, k_h)] \geq 0, \quad (8)$$

$$(1 - \delta)(u^{dd} - u^{cd}) + \delta[\mathbb{P}(z > k_h \mid dd) - \mathbb{P}(z > k_h \mid cd)][v_C(k_l, k_h) - v_P(k_l, k_h)] \geq 0. \quad (9)$$

The first inequality deters defection in the cooperation state. The second deters cooperation in the punishment state: a one-period cooperative deviation sacrifices the current mutual-defection advantage but makes a high signal, and hence a return to cooperation, more likely. Importantly, the next proposition shows that, for any given pair of thresholds, the continuation values are uniquely determined and that, as expected, the value in the cooperation state is higher than in the punishment state.

Proposition 2. *For every finite (k_l, k_h) , (6)–(7) have a unique solution. Moreover,*

$$v_C(k_l, k_h) - v_P(k_l, k_h) = \frac{(1 - \delta)(u^{cc} - u^{dd})}{1 - \delta[\mathbb{P}(z > k_l \mid cc) - \mathbb{P}(z > k_h \mid dd)]} > 0. \quad (10)$$

Define the equilibrium-feasible and pairwise p -dominant threshold sets by

$$\mathcal{F} = \{(k_l, k_h) \in \mathbb{R}^2 : (8) \text{ and } (9) \text{ hold}\},$$

and by $\mathcal{F}^p \subseteq \mathcal{F}$ the set of thresholds such that the associated strategies are pairwise p -dominant. The corresponding payoff frontiers are

$$v_F^* = \sup_{(k_l, k_h) \in \mathcal{F}} v_C(k_l, k_h), \quad v_F^p = \sup_{(k_l, k_h) \in \mathcal{F}^p} v_C(k_l, k_h).$$

Theorem 3. *Maintain the assumptions of Theorem 1 and fix $p \in (0, 1)$. If $\mathcal{F}$ and $\mathcal{F}^p$ are nonempty, then*

$$v_F^* = \bar{v}^{\text{PPE}}, \quad v_F^p < v_F^*.$$

The first equality replicates the threshold-Grim result in Aoyagi and Fréchette (2009) and is attained within the extended class at $k_h = +\infty$, where the public-forgiving strategy collapses to threshold Grim.

IV EMPIRICAL ANALYSIS

Aoyagi and Fréchette (2009) implement an indefinitely repeated game with one-dimensional public monitoring. In each period, subjects choose between cooperation, L , and defection, H . Expected stage payoffs are

	L	H	
L	25, 25	15, 28	(11)
H	28, 15	16, 16.	

Aoyagi and Fréchette (2009) implement an additive public-monitoring structure in which, following action profile a , the public signal is $z = s(a) + x$, with $s(LL) = 20$, $s(LH) = s(HL) = 18$ and $s(HH) = 16$. For $\beta > 0$, the shock x follows a mean-zero Laplace distribution with scale parameter β ,

$$f_\beta(x) = \frac{1}{2\beta} \exp\left(-\frac{|x|}{\beta}\right), \quad F_\beta(x) = \begin{cases} \frac{1}{2}e^{x/\beta}, & x < 0, \\ 1 - \frac{1}{2}e^{-x/\beta}, & x \geq 0, \end{cases} \quad (12)$$

so that $\text{Var}(x) = 2\beta^2$. The four main treatments set $\beta \in \{0, 1, 4, 10\}$, with higher β corresponding to noisier monitoring.⁵ The Laplace monitoring technologies are ordered by Blackwell informativeness, so that a treatment with higher β is a garbling of one with lower β . The comparative-statics result of Kandori (1992) therefore applies: the pure-strategy sequential-equilibrium payoff set contracts as monitoring becomes noisier. This is consistent with the main result in Aoyagi and Fréchette (2009), which shows that the maximal equilibrium value declines as monitoring becomes noisier and that subjects' behavior closely tracks the corresponding theoretical benchmarks.

⁵For $\beta = 0$, monitoring is deterministic: $x = 0$ almost surely, so $F_0(t) = \mathbf{1}\{t \geq 0\}$. There are two sessions for each treatment, $\beta \in \{0, 1, 4, 10\}$, with continuation probability $\delta = 0.9$.

IV(i) EMPIRICAL STRATEGY ESTIMATION

Following Aoyagi and Fréchette (2009), I restrict attention to stable behavior by excluding the first two supergames and any supergame beginning before 20 session periods have elapsed, and I drop subject-supergames containing missing signals. The empirical model is a finite mixture estimated separately for each noise treatment and includes the three strategy classes described in the previous section: AllD, public threshold-Grim, and public-forgiving strategies. AllD prescribes H in every period, up to a small cooperation-error probability. Public threshold-Grim prescribes L initially and switches permanently to punishment after a signal $z \leq a$, allowing for action errors during punishment. Public-forgiving strategies prescribe L initially, enter punishment after $z \leq k_l$, and return to cooperation after $z > k_h$.

I use subjects' observed choices over time to estimate the prevalence of the three behavioral patterns in each treatment and how the cooperative types respond to public signals. For each subject, the fitted model yields posterior probabilities of belonging to each strategy class, conditional on the estimated strategy thresholds. I also report a hard classification that assigns each subject to the class with the highest posterior probability, thereby collapsing posterior uncertainty into a discrete type assignment.⁶ Table 1 reports the estimation results.

β	Subjects	Mixture-estimated share (%)			Hard share (%)		
		AllD	Grim	Forgiving	AllD	Grim	Forgiving
0	38	12.4	10.0	77.6	5.3 (2)	13.2 (5)	81.6 (31)
1	42	8.2	5.6	86.2	4.8 (2)	4.8 (2)	90.5 (38)
4	40	8.1	13.2	78.7	5.0 (2)	12.5 (5)	82.5 (33)
10	46	16.1	7.0	76.9	13.0 (6)	4.3 (2)	82.6 (38)

Notes: Mixture shares average posterior membership probabilities and sum to 100 percent within treatment. Hard shares assign each subject to the class with the largest posterior; counts are in parentheses. The estimation sample excludes the one-shot and infinite-noise controls.

Table 1: Estimated strategy shares

Posterior mass is concentrated on public-forgiving strategies in every treatment, accounting for roughly three quarters or more of the estimated mixture share. The hard classifications tell a similar story, assigning more than 80 percent of subjects to the forgiving type in all treatments. There is little systematic movement in strategy composition as monitoring becomes noisier, although at the highest noise level, $\beta = 10$,

⁶Details of the estimation are reported in Appendix F.

some mass shifts toward AllD. In this treatment, cooperation cannot be sustained in equilibrium.

To map the theoretical predictions about value into the data, for each subject hard-classified as threshold-Grim or public forgiving, I match the subject with a hypothetical opponent using the same strategy class and the same posterior-mean threshold or threshold pair. The resulting quantities are *symmetric self-play values*: they measure the value each estimated strategy would generate if the opponent followed the same strategy, providing the empirical analogue of the theoretical self-play value $v(\sigma)$. Importantly, I do not impose the equilibrium incentive constraints when computing these values, so an estimated strategy need not itself constitute an S -PPE. Figure 1 reports the resulting self-play values for subjects classified as using threshold-Grim strategies, while Figure 2 reports the corresponding values for subjects classified as using public-forgiving strategies.

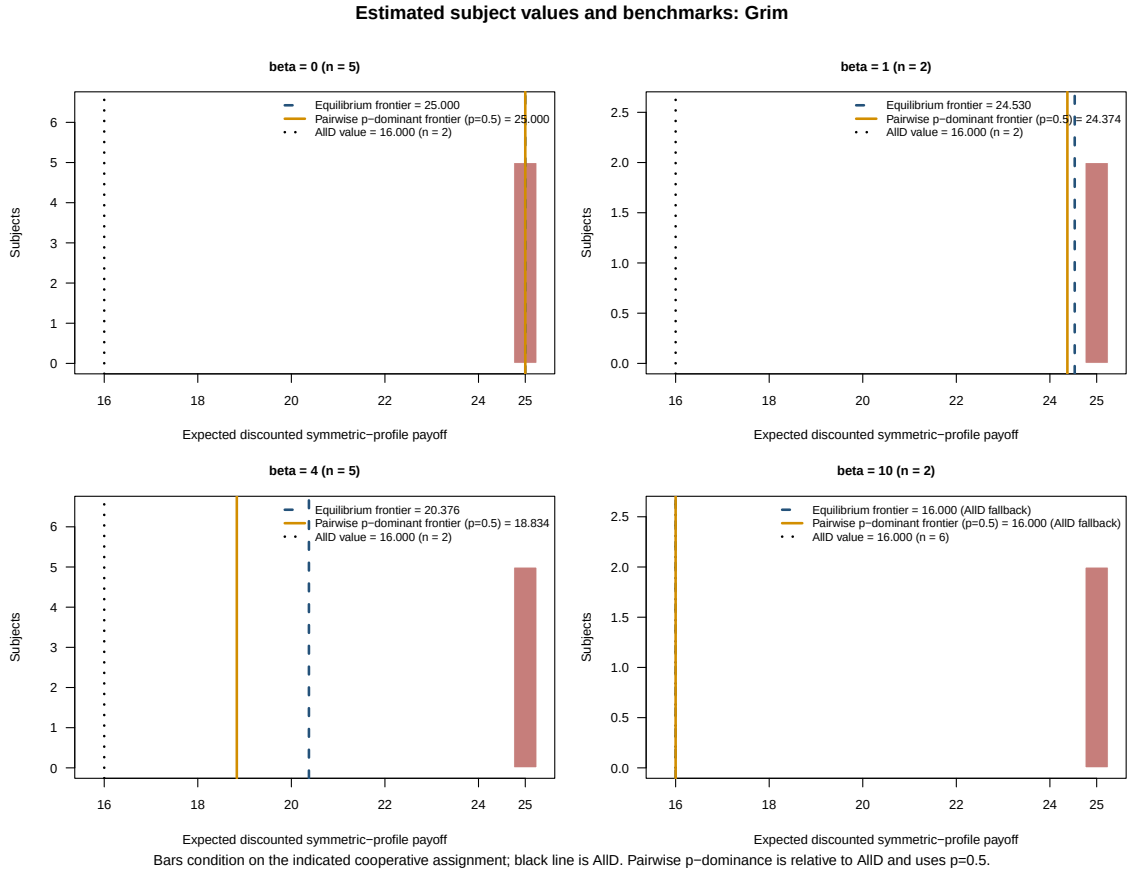


Figure 1: Estimated subject-level self-play values for hard-assigned threshold-Grim subjects. Blue dashed lines show the exact equilibrium frontier, and gold solid lines impose, in addition, pairwise p -dominance at $p = 1/2$, relative to AllD. Black dotted lines mark the AllD value of 16 and report the number of hard-assigned AllD subjects.

Estimated subject values and benchmarks: Public forgiving

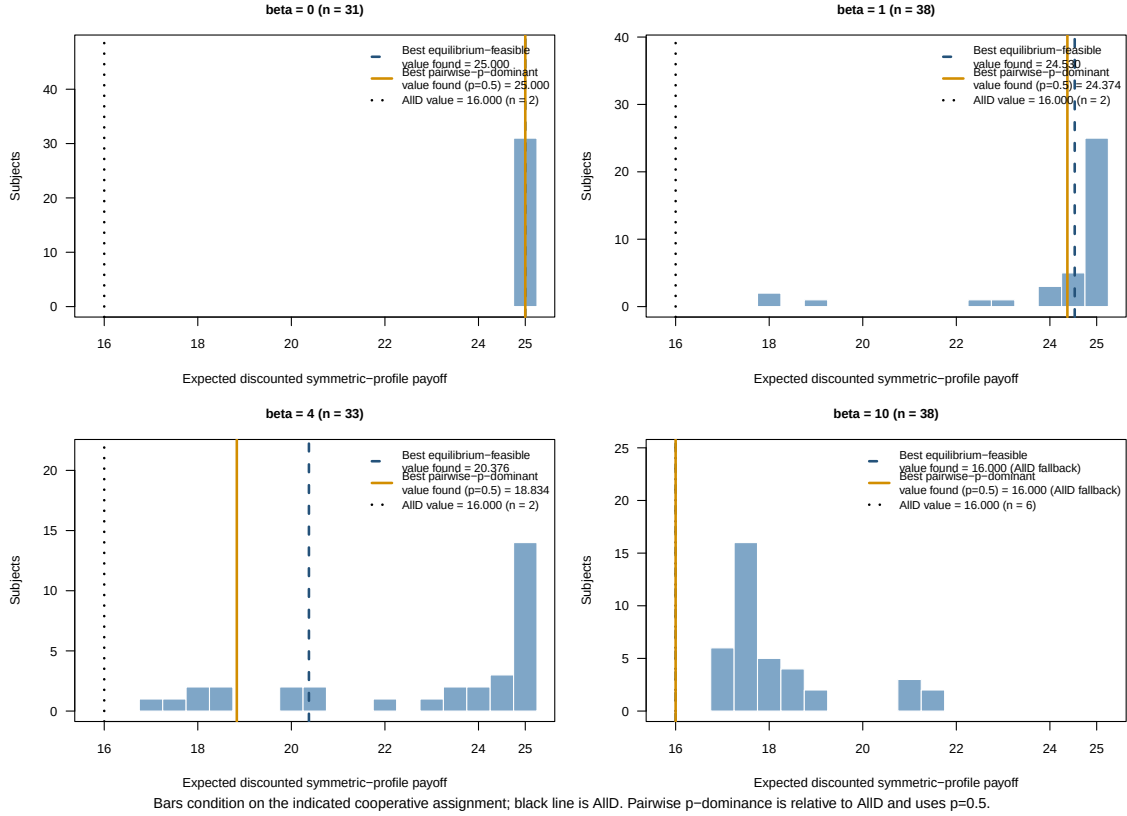


Figure 2: Estimated subject-level self-play values for hard-assigned public-forgiving subjects. Blue dashed lines show the best equilibrium-feasible values; gold solid lines impose, in addition, pairwise p -dominance at $p = 1/2$, relative to AllD. Black dotted lines mark the AllD value of 16 and report the number of hard-assigned AllD subjects.

Conditional on adopting a cooperative strategy, many subjects choose strategies that preserve a high degree of self-play efficiency, at times even failing the equilibrium incentive constraints. Together, these patterns provide suggestive evidence that, conditional on cooperating, subjects choose thresholds that are more closely related to maximizing the value of cooperation than to limiting exposure to strategic risk. Pairwise p -dominance therefore appears to have limited power to explain the choice among cooperative strategies under imperfect monitoring. This contrasts with evidence under perfect monitoring, where strategic risk helps predict the extensive-margin decision of whether to adopt a cooperative strategy at all.

A word of caution is needed in interpreting these results. The theory characterizes the strategies that can be sustained in equilibrium and under pairwise p -dominance, whereas the experimental treatments do not vary strategic risk in isolation. Changes in

monitoring noise affect both the attainable value of cooperation and the informativeness of public signals. A more direct experimental test would therefore require variation that separately shifts the value–risk trade-off, allowing the role of strategic risk in the choice among cooperative strategies to be identified more cleanly. The following proposition shows that such an experiment is possible.

Proposition 3 (Payoff variation and strategic risk). *Maintain Assumption 2 and fix the discount factor δ , the monitoring technology, and u^{cc} , u^{dc} , and u^{dd} . If u^{cd} decreases then, in both the threshold-Grim and public-forgiving classes, the S-PPE equilibrium value ($\bar{v}^{\text{PPE}}$) is unchanged, whereas the payoff-exposure condition for pairwise p -dominance becomes strictly harder to satisfy.*

V CONCLUSION

This paper formalizes the trade-off between strategic risk and the value of cooperation in repeated Prisoner’s Dilemmas with imperfect public monitoring. Under perfect monitoring, grim trigger simultaneously delivers the highest value of cooperation and the greatest strategic safety. Under imperfect public monitoring, because unfavorable signals occur on the equilibrium path, strategies that are more tolerant of bad signals can sustain cooperation for longer and generate higher values, but they also leave a player more exposed to opportunistic behavior. Formally, I show that under standard assumptions, any nondegenerate belief that the opponent may always defect implies that the maximal payoff attainable in a pairwise p -dominant symmetric PPE is strictly below the unconstrained symmetric-PPE frontier.

I then operationalize this mechanism within economically interpretable classes of threshold strategies and relate it to the experimental data. A reanalysis of the data from Aoyagi and Fr  chette (2009) provides suggestive evidence that subjects favor strategies that preserve the value of cooperation, rather than strategically safer, pairwise p -dominant equilibria. The punchline is that pairwise p -dominance appears to have limited power to explain the choice among cooperative strategies under imperfect monitoring. This contrasts with evidence from perfect-monitoring environments, where strategic risk appears to matter for the initial choice of whether to cooperate, but is silent on the selection among cooperative strategies.

Appendix

A Preliminary Results and the Payoff–Exposure Representation

For the proofs, define

$$C(k) = \mathbb{P}(z \geq k \mid cc), \quad D(k) = \mathbb{P}(z \geq k \mid dc), \quad \Lambda_\delta(k) = \frac{1 - \delta D(k)}{1 - \delta C(k)},$$

$$v_G(k) = u^{dd} + \frac{(1 - \delta)(u^{cc} - u^{dd})}{1 - \delta C(k)}, \quad \ell = \frac{u^{dc} - u^{dd}}{u^{cc} - u^{dd}}, \quad j(p) = \frac{\chi(p)}{u^{cc} - u^{dd}},$$

$$K(\delta) = \{k : \Lambda_\delta(k) \geq \ell\}, \quad K^p(\delta) = \{k : \Lambda_\delta(k) \geq j(p)\},$$

and

$$k^*(\delta) = \min K(\delta), \quad k^p(\delta) = \min K^p(\delta).$$

Arguments are suppressed when no confusion can arise.

Lemma 2 (The all-defect equilibrium). AllD $\in \mathcal{S}_\delta$ and $v(\text{AllD}) = u^{dd}$.

Proof. After every public history, if the opponent chooses d , current defection yields u^{dd} and current cooperation yields $u^{cd} < u^{dd}$. A one-period deviation does not change the future actions prescribed by AllD. The public one-shot-deviation principle therefore implies that (AllD, AllD) is a PPE. Its normalized payoff is u^{dd} . $\square$

Proof of Lemma 1. Fix σ . Under (σ, AllD) , the action profile after public history h is $(\sigma(h), d)$; under (AllD, σ) , it is $(d, \sigma(h))$. If $\sigma(h) = d$, both profiles are dd . If $\sigma(h) = c$, the profiles are cd and dc , which induce the same signal distribution by Assumption 2. Induction on the length of public histories implies that the two cross-matches induce the same law over public histories.

Under this common law, σ cooperates with normalized discounted frequency $x(\sigma)$. Substitution into (2) and division by $p > 0$ give

$$v(\sigma) - u^{dd} \geq \left[u^{dc} - u^{dd} + \frac{1-p}{p}(u^{dd} - u^{cd}) \right] x(\sigma).$$

This is (4). $\square$

B Proof of the General Strict Top-Frontier Theorem

Proof of Theorem 1. Fix $p \in (0, 1)$ and write $\bar{v} = \bar{v}^{\text{PPE}}$. Compactness of $\mathcal{E}_\delta$ implies that some $\sigma^* \in \mathcal{S}_\delta$ attains $\bar{v}$. By hypothesis, $\bar{v} > u^{dd}$.

Step 1: a threshold Grim strategy attains the top payoff. Fix any top-payoff strategy σ . It prescribes c initially. If it prescribed d , every continuation value would be at most $\bar{v}$, and

$$\bar{v} \leq (1 - \delta)u^{dd} + \delta\bar{v},$$

contradicting $\bar{v} > u^{dd}$.

Let $W(z)$ be the symmetric continuation payoff after first-period signal z following cc . Every symmetric PPE continuation value is at most $\bar{v}$. It is also at least u^{dd} : a player can deviate to defection at every future history, obtaining at least u^{dd} in every period because $u^{dc} > u^{dd}$. Hence

$$y(z) = \frac{W(z) - u^{dd}}{\bar{v} - u^{dd}} \in [0, 1].$$

Set

$$c = \int_Z y(z)p(z \mid cc) dz, \quad d = \int_Z y(z)p(z \mid dc) dz.$$

The payoff equation and cooperative-state deviation constraint are

$$\bar{v} = (1 - \delta)u^{cc} + \delta\{u^{dd} + (\bar{v} - u^{dd})c\}, \tag{13}$$

$$\bar{v} \geq (1 - \delta)u^{dc} + \delta\{u^{dd} + (\bar{v} - u^{dd})d\}. \tag{14}$$

Choose a cutoff k such that

$$C(k) = \int_k^{\sup Z} p(z \mid cc) dz = c,$$

using the endpoint convention for $c \in \{0, 1\}$. The strict likelihood-ratio ordering implies

$$\int_Z y(z)p(z \mid dc) dz \geq D(k). \tag{15}$$

For an interior cutoff, define

$$h(z) = y(z) - \mathbf{1}\{z \geq k\}, \quad L(z) = \frac{p(z \mid cc)}{p(z \mid dc)}.$$

Then $\int h(z)p(z \mid cc) dz = 0$, L is strictly increasing, $h \geq 0$ below k , and $h \leq 0$ above k .

Therefore

$$\int_Z h(z)[L(z) - L(k)]p(z \mid dc) dz \leq 0.$$

The first term after expansion is zero, because $\int hp(\cdot \mid cc) = 0$. Since $L(k) > 0$, the remaining term gives $\int hp(\cdot \mid dc) \geq 0$, which is (15). Strict monotonicity also shows that equality requires $y(z) = \mathbf{1}\{z \geq k\}$ almost everywhere. Endpoint cases follow by continuity.

Define a two-state public automaton directly. In its cooperative state, both players choose c ; after $z \geq k$ the automaton returns to that same state, and after $z < k$ it enters the absorbing AllD state. Its cooperative-state Bellman equation is (13) with $y = \mathbf{1}\{z \geq k\}$, so its value is $\bar{v}$. This replacement leaves the cooperative continuation probability unchanged and, by (15), weakly lowers the continuation payoff from a one-period defection. Hence (14) remains satisfied. The absorbing state is a PPE by Lemma 2; the public one-shot-deviation principle therefore establishes that the constructed stationary threshold-Grim strategy is a symmetric PPE attaining $\bar{v}$.

For any threshold-Grim strategy,

$$v_G(k) - u^{dd} = \frac{(1 - \delta)(u^{cc} - u^{dd})}{1 - \delta C(k)},$$

and the cooperative-state incentive constraint is

$$\Lambda_\delta(k) = \frac{1 - \delta D(k)}{1 - \delta C(k)} \geq \ell.$$

Thus $K(\delta)$ is nonempty. Continuity and full support imply $\Lambda_\delta(k) \rightarrow 1$ at both endpoint cutoffs. Because $\ell > 1$, $K(\delta)$ is closed and bounded away from both endpoints. Let $k^* = \min K(\delta)$. Positivity of $p(\cdot \mid cc)$ makes $v_G(k)$ strictly decreasing in k , so $v_G(k^*) = \bar{v}$. Continuity and minimality imply

$$\Lambda_\delta(k^*) = \ell. \tag{16}$$

Step 2: identify the least exposure among top-payoff PPEs. Compactness of $\mathcal{E}_\delta$ implies that

$$x^* = \min\{x : (\bar{v}, x) \in \mathcal{E}_\delta\}$$

exists. For any top-payoff strategy, compare (13) with the corresponding equation for k^* . This gives $c = C(k^*)$. Comparing (14) with the binding threshold constraint gives $d \leq D(k^*)$. The Neyman–Pearson inequality gives the reverse inequality. Equality

therefore holds, and strict likelihood-ratio ordering implies

$$y(z) = \mathbf{1}\{z \geq k^*\} \quad \text{almost everywhere.}$$

Every top-payoff PPE consequently continues with payoff $\bar{v}$ after $z \geq k^*$ and payoff u^{dd} after $z < k^*$, up to null sets.

If a symmetric PPE continuation τ has $v(\tau) = u^{dd}$, then $x(\tau) = 0$. Otherwise Lemma 1 would give

$$V_{\text{AllD},\tau} = u^{dd} + (u^{dc} - u^{dd})x(\tau) > v(\tau),$$

contradicting the best-response property of τ . Let σ_z denote the continuation strategy after z . Any top-payoff strategy obeys

$$x(\sigma) = (1 - \delta) + \delta \int_Z x(\sigma_z) p(z \mid cd) dz \geq (1 - \delta) + \delta D(k^*) x^*.$$

Taking the minimum over top-payoff strategies yields

$$x^* \geq \frac{1 - \delta}{1 - \delta D(k^*)}.$$

The threshold-Grim strategy at k^* attains equality, because its exposure solves $x_G = (1 - \delta) + \delta D(k^*) x_G$. Hence

$$x^* = \frac{1 - \delta}{1 - \delta D(k^*)}. \quad (17)$$

Step 3: no top-payoff PPE is pairwise p -dominant. Among top-payoff equilibria, (4) is easiest to satisfy at exposure x^* . Substituting the top Grim value and (17), the necessary condition is

$$\frac{1 - \delta D(k^*)}{1 - \delta C(k^*)} \geq \frac{\chi(p)}{u^{cc} - u^{dd}}.$$

By (16), the left side is ℓ , whereas

$$\frac{\chi(p)}{u^{cc} - u^{dd}} - \ell = \frac{1 - p}{p} \frac{u^{dd} - u^{cd}}{u^{cc} - u^{dd}} > 0.$$

Strictness uses $p < 1$ and $u^{dd} > u^{cd}$. Thus no point in $\mathcal{E}_\delta$ with payoff $\bar{v}$ satisfies the dominance half-space.

Finally, AllD ensures that the pairwise p -dominant set is nonempty. Its payoff-exposure image is the intersection of compact $\mathcal{E}_\delta$ with the closed half-space (4); its maximum payoff is therefore attained. Because that compact set contains no point with payoff $\bar{v}$, its maximum is strictly below $\bar{v}$. $\square$

C Characterization of Threshold-Grim Strategies

Proof of Proposition 1. In the cooperative state, symmetric play gives

$$v_G(k) = (1 - \delta)u^{cc} + \delta C(k)v_G(k) + \delta[1 - C(k)]u^{dd},$$

which solves to (5). A one-period defection yields

$$(1 - \delta)u^{dc} + \delta D(k)v_G(k) + \delta[1 - D(k)]u^{dd}.$$

Using (5), cooperation is optimal if and only if

$$\frac{1 - \delta D(k)}{1 - \delta C(k)} \geq \frac{u^{dc} - u^{dd}}{u^{cc} - u^{dd}} = \ell.$$

In the absorbing punishment state, defection is strictly optimal because $u^{dd} > u^{cd}$. The public one-shot-deviation principle gives necessity and sufficiency.

Against AllD, Grim exposure solves

$$x_G(k) = (1 - \delta) + \delta D(k)x_G(k),$$

so

$$x_G(k) = \frac{1 - \delta}{1 - \delta D(k)}. \tag{18}$$

Substituting (5) and (18) into (4) gives

$$\frac{1 - \delta D(k)}{1 - \delta C(k)} \geq \frac{\chi(p)}{u^{cc} - u^{dd}} = j(p).$$

At $p = 1/2$,

$$j(1/2) = \frac{u^{dc} - u^{dd} + u^{dd} - u^{cd}}{u^{cc} - u^{dd}} = \frac{u^{dc} - u^{cd}}{u^{cc} - u^{dd}}.$$

□

Lemma 3 (Interval geometry). *For any target $T > 1$, define*

$$K_T = \{k : \Lambda_\delta(k) \geq T\}.$$

If $p(z \mid dc)/p(z \mid cc)$ is decreasing and the densities are continuous and positive, then K_T is a possibly empty compact interval. If nonempty, its lower endpoint satisfies $\Lambda_\delta(k_T) = T$.

Proof. The inequality $\Lambda_\delta(k) \geq T$ is equivalent to

$$W_T(k) := TC(k) - D(k) \geq \frac{T-1}{\delta}.$$

Write

$$W_T(k) = \int_k^{\sup Z} \left[T - \frac{p(z \mid dc)}{p(z \mid cc)} \right] p(z \mid cc) dz.$$

The bracketed term is increasing in z . Hence its sign changes at most once. Since

$$W'_T(k) = - \left[T - \frac{p(k \mid dc)}{p(k \mid cc)} \right] p(k \mid cc),$$

W_T first weakly increases and then weakly decreases. Every superlevel set is therefore an interval. Continuity makes it closed.

At the lower endpoint of Z , $C, D \rightarrow 1$, so $\Lambda_\delta \rightarrow 1 < T$; at the upper endpoint, $C, D \rightarrow 0$, so the same limit is one. Thus the superlevel set is bounded away from both endpoints and is compact. At a finite lower endpoint of a nonempty set, the inequality must bind by continuity. $\square$

Proof of Theorem 2. Apply Lemma 3 to $T = \ell$ and $T = j(p)$. Equation (5) and positivity of the cc density imply that $v_G(k)$ is strictly decreasing in k . The lower endpoints therefore attain the two class-specific value frontiers.

For $p < 1$,

$$j(p) - \ell = \frac{1-p}{p} \frac{u^{dd} - u^{cd}}{u^{cc} - u^{dd}} > 0.$$

Thus $K^p(\delta) \subseteq K(\delta)$, and every point of $K^p(\delta)$ satisfies the equilibrium inequality strictly. At the lower equilibrium endpoint k^* , $\Lambda_\delta(k^*) = \ell < j(p)$, so $k^p > k^*$. Strictly decreasing v_G gives $v_G(k^p) < v_G(k^*)$. $\square$

Corollary 1 (Belief comparative statics). *At a differentiable lower endpoint with $\Lambda'_\delta(k^p) > 0$,*

$$\frac{dk^p}{dp} = - \frac{u^{dd} - u^{cd}}{p^2(u^{cc} - u^{dd})\Lambda'_\delta(k^p)} < 0.$$

Consequently, the maximal threshold-Grim pairwise p -dominant value increases with p .

Proof. The lower endpoint solves $\Lambda_\delta(k^p) = j(p)$, where

$$j'(p) = - \frac{u^{dd} - u^{cd}}{p^2(u^{cc} - u^{dd})} < 0.$$

Implicit differentiation gives the displayed expression. Because v_G is strictly decreasing

in its cutoff, $dv_G(k^p)/dp > 0$. □

D Characterization of Two-State Public-Forgiving Strategies

For compact appendix notation only, set

$$A = \mathbb{P}(z > k_l \mid cc), \quad B = \mathbb{P}(z > k_h \mid dd), \quad D = \mathbb{P}(z > k_l \mid dc), \quad E = \mathbb{P}(z > k_h \mid cd).$$

All four quantities depend on the thresholds and monitoring technology.

Proof of Proposition 2. The Bellman system can be written

$$\begin{pmatrix} 1 - \delta A & -\delta(1 - A) \\ -\delta B & 1 - \delta(1 - B) \end{pmatrix} \begin{pmatrix} v_C \\ v_P \end{pmatrix} = (1 - \delta) \begin{pmatrix} u^{cc} \\ u^{dd} \end{pmatrix}.$$

Its determinant is

$$\begin{aligned} \Delta_v &= [1 - \delta A][1 - \delta(1 - B)] - \delta^2 B(1 - A) \\ &= (1 - \delta)[1 - \delta(A - B)]. \end{aligned}$$

Since $A - B \in [-1, 1]$ and $\delta \in (0, 1)$, $1 - \delta(A - B) \geq 1 - \delta > 0$. Thus $\Delta_v > 0$, proving existence and uniqueness.

Subtracting the two Bellman equations gives

$$v_C - v_P = (1 - \delta)(u^{cc} - u^{dd}) + \delta(A - B)(v_C - v_P).$$

Solving proves (10). The individual values can also be written as

$$v_C = u^{cc} - \frac{\delta(1 - A)(u^{cc} - u^{dd})}{1 - \delta(A - B)}, \tag{19}$$

$$v_P = u^{dd} + \frac{\delta B(u^{cc} - u^{dd})}{1 - \delta(A - B)}. \tag{20}$$

In state C , a one-period defection gives current payoff u^{dc} and returns to C with probability D . Subtracting its value from the prescribed value gives

$$(1 - \delta)(u^{cc} - u^{dc}) + \delta(A - D)(v_C - v_P),$$

which is (8). In state P , a one-period cooperation gives u^{cd} and returns to C with

probability E . The gain from following prescribed defection is

$$(1 - \delta)(u^{dd} - u^{cd}) + \delta(B - E)(v_C - v_P),$$

which is (9). These are the only two public states and the only one-period action deviations. The public one-shot-deviation principle therefore makes the two inequalities necessary and sufficient. $\square$

For use in the comparative statics below, exposure against AllD from the cooperative state is

$$x_C(k_l, k_h) = \frac{1 - \delta + \delta B}{1 - \delta(D - B)}. \quad (21)$$

By Lemma 1, an equilibrium-feasible pair is pairwise p -dominant if and only if

$$v_C(k_l, k_h) - u^{dd} \geq \chi(p)x_C(k_l, k_h). \quad (22)$$

Proof of Theorem 3. The proof of Theorem 1 shows that a threshold-Grim strategy with cutoff $k^*(\delta)$ attains $\bar{v}^{\text{PPE}}$.

Because $\mathcal{F}^p$ is nonempty, fix a finite pair $(\hat{k}_l, \hat{k}_h) \in \mathcal{F}^p$. At a fixed entry threshold, (10) converges as $k_h \rightarrow +\infty$ to the continuation-value difference under threshold Grim. For every finite k_h , full support implies $\mathbb{P}(z > k_h \mid dd) > 0$, so the forgiving continuation-value difference is strictly smaller. Strict likelihood-ratio ordering also implies

$$\mathbb{P}(z > k_l \mid cc) - \mathbb{P}(z > k_l \mid dc) > 0.$$

It follows from (8) that the threshold-Grim cooperation constraint at $\hat{k}_l$ holds strictly. Hence $\hat{k}_l$ lies in the interior of the feasible threshold-Grim interval, whose lower endpoint is $k^*(\delta)$.

Choose a sequence $k_{l,n} \downarrow k^*(\delta)$ from the interior of that interval. For each n , the threshold-Grim cooperation constraint at $k_{l,n}$ is strict. As $k_h \rightarrow +\infty$, the cooperative-state constraint of the forgiving strategy converges to this strict inequality, while its punishment-state constraint converges to

$$(1 - \delta)(u^{dd} - u^{cd}) > 0.$$

Continuity therefore allows a finite $k_{h,n}$, with $k_{h,n} \rightarrow +\infty$, such that $(k_{l,n}, k_{h,n}) \in \mathcal{F}$. The corresponding values satisfy

$$v_C(k_{l,n}, k_{h,n}) \longrightarrow v_C(k^*(\delta)) = \bar{v}^{\text{PPE}}.$$

Since every public-forgiving PPE is also a symmetric public PPE, the reverse inequality is immediate. Thus $v_F^* = \bar{v}^{\text{PPE}}$.

Finally, every pairwise p -dominant public-forgiving strategy belongs to $\mathcal{S}_\delta^p$. Therefore, by Theorem 1,

$$v_F^p \leq \bar{v}^p < \bar{v}^{\text{PPE}} = v_F^*,$$

which proves the strict separation. $\square$

E Proof of the Payoff-Variation Proposition

Proof of Proposition 3. Under symmetric self-play, the two players use the same public strategy after every public history and therefore choose the same action. Consequently, self-play generates only the action profiles cc and dd , never cd . The self-play value of any fixed threshold-Grim or public-forgiving strategy is therefore independent of u^{cd} .

Now lower u^{cd} to $\tilde{u}^{cd} < u^{cd}$, holding the strategy, discount factor, and monitoring technology fixed. The exposure $x(\sigma)$ against AllD is unchanged. By Lemma 1, the payoff-exposure condition for pairwise p -dominance is

$$v(\sigma) - u^{dd} \geq \chi(p)x(\sigma), \quad \chi(p) = u^{dc} - u^{dd} + \frac{1-p}{p}(u^{dd} - u^{cd}).$$

After the decrease,

$$\tilde{\chi}(p) - \chi(p) = \frac{1-p}{p}(u^{cd} - \tilde{u}^{cd}) > 0.$$

Every strategy in either class begins in the cooperation state, so its exposure against AllD satisfies $x(\sigma) \geq 1 - \delta > 0$. Thus the right-hand side of the dominance inequality rises strictly, while its left-hand side is unchanged. The condition is therefore strictly harder to satisfy.

For threshold-Grim strategies, u^{cd} enters neither the cooperation-state incentive constraint nor the self-play value; lowering it only reinforces defection in the absorbing punishment state. Hence the equilibrium-feasible set is unchanged. Under the additional assumptions of Theorem 2, if the pairwise p -dominant cutoff sets before and after the change are both nonempty, the higher value of $\chi(p)$ raises the minimal admissible cutoff and, because $v_G(k)$ is strictly decreasing in k , lowers the maximal pairwise p -dominant value in this class.

For public-forgiving strategies, lowering u^{cd} also increases the left-hand side of the

punishment-state incentive constraint (9) by

$$(1 - \delta)(u^{cd} - \tilde{u}^{cd}) > 0.$$

It therefore relaxes equilibrium feasibility while tightening the payoff-exposure condition for pairwise p -dominance. Because these effects work in opposite directions, they provide no unambiguous comparative static for the maximal pairwise p -dominant public-forgiving value without further restrictions. $\square$

F Empirical and Computational Appendix

F(i) Data construction

A summary of the numbers of subjects and observations is shown in table 2.

Table 2: Mixture-estimation sample

β	Subjects	Usable total choices	Usable noninitial choices
0	38	2,150	1,908
1	42	1,296	1,112
4	40	1,480	1,360
10	46	1,474	1,258

Notes: The sample drops the first two supergames and any additional supergame that begins before 20 session periods have elapsed. A subject-supergame is removed if any public signal is missing.

F(ii) Likelihood construction

Each treatment is estimated separately. Its eleven-dimensional parameter vector is

$$\theta = (\eta_G, \eta_F, \alpha_G, \lambda_G, \alpha_F, \lambda_F, \Delta_F, \psi, \lambda_\omega, \eta_d, \eta_c).$$

With AllD as the base category, $D_\pi = 1 + e^{\eta_G} + e^{\eta_F}$ and

$$(\pi_A, \pi_G, \pi_F) = D_\pi^{-1}(1, e^{\eta_G}, e^{\eta_F}).$$

The signal slopes and random-effect standard deviation are $\gamma_s = e^{\lambda_s} > 0$ and $\omega = e^{\lambda_\omega} > 0$. The two error probabilities are constrained to $(0.001, 0.25)$ by

$$\epsilon_d = 0.001 + 0.249\Lambda(\eta_d), \quad \epsilon_c = 0.001 + 0.249\Lambda(\eta_c),$$

where Λ is the logistic CDF. Here ϵ_d is the cooperation probability for AllD in every period and for Grim once its absorbing punishment state is active; ϵ_c is the period-one defection probability for both cooperative classes.

Let ζ_i be subject i 's first-period defection rate in the estimation sample and let $\xi_i \sim N(0, 1)$. Conditional on ξ_i , the class-specific thresholds are

$$a_{iG}(\xi_i) = \frac{\alpha_G + \psi\zeta_i + \omega\xi_i}{\gamma_G}, \quad (23)$$

$$k_{l,iF}(\xi_i) = \frac{\alpha_F + \psi\zeta_i + \omega\xi_i}{\gamma_F}, \quad k_{h,iF}(\xi_i) = k_{l,iF}(\xi_i) + \frac{\Delta_F}{\gamma_F}. \quad (24)$$

Each supergame begins in the cooperative state. For a noninitial choice following public signal z , a cooperative-state Grim subject chooses cooperation with probability $\Phi(\gamma_G[z - a_{iG}])$; once the absorbing punishment state is active, that probability is ϵ_d . A forgiving subject uses $\Phi(\gamma_F[z - k_{l,iF}])$ when the active state is cooperative and $\Phi(\gamma_F[z - k_{h,iF}])$ when it is punishment.

The public-state recursion is deterministic conditional on the public signal and random-effect node. Grim remains in cooperation only if it was already there and $z > a_{iG}$. Forgiving is in cooperation next period if z exceeds the threshold active in the current state.

Let $y_{igt} \in \{0, 1\}$ denote cooperation by subject i in period t of supergame g , and let $q_{igst}(\xi)$ be the corresponding class- and node-specific cooperation probability, including the period-one probabilities described above. Conditional on the observed public-signal histories,

$$\ell_{is}(\xi) = \sum_g \sum_t [y_{igt} \log q_{igst}(\xi) + (1 - y_{igt}) \log \{1 - q_{igst}(\xi)\}].$$

For twelve Golub–Welsch Gauss–Hermite nodes and weights $\{(\xi_m, w_m)\}_{m=1}^{12}$, transformed so that $\sum_m w_m = 1$ under a standard normal expectation, the integrated type likelihood and mixture contribution are

$$L_{is} = \sum_{m=1}^{12} w_m \exp\{\ell_{is}(\xi_m)\}, \quad L_i = \pi_A L_{iA} + \pi_G L_{iG} + \pi_F L_{iF}.$$

Both the quadrature and mixture sums are evaluated with log-sum-exp stabilization.

The objective $\sum_i \log L_i$ is a conditional behavioral-response composite likelihood: it conditions on the realized public-signal and supergame histories and treats subject choice innovations, latent classes, and random effects as independent across subjects. It does not model the signal density or the joint session process and is not a full joint likelihood of actions and signals. The class weights reported below are therefore model-based

empirical-Bayes probabilities conditional on the estimated parameters, not fully Bayesian posteriors.

F(iii) Posterior classification and thresholds

Posterior class probabilities are

$$\mathbb{P}(s \mid \text{data}_i) = \frac{\pi_s L_{is}}{\sum_{q \in \{A, G, F\}} \pi_q L_{iq}}.$$

Hard classification selects the largest posterior. Conditional on a cooperative class, quadrature-node posteriors produce an empirical-Bayes mean and variance for $\omega \xi_i$.

F(iv) Value and incentive calculations

For hard-assigned Grim, the counterfactual uses

$$\mathbb{P}(C \rightarrow C) = 1 - F_\beta(a_i - 20)$$

and (5). For hard-assigned forgiving, the transition matrix is

$$\begin{pmatrix} 1 - F_\beta(k_{l,i} - 20) & F_\beta(k_{l,i} - 20) \\ 1 - F_\beta(k_{h,i} - 16) & F_\beta(k_{h,i} - 16) \end{pmatrix},$$

and the value solves the two-state Bellman system. AllD receives 16. The calculations omit estimated action errors and do not integrate over the posterior distribution of thresholds.

F(v) Benchmark computation

The threshold-Grim benchmarks solve the one-dimensional cutoff problem by a dense deterministic search followed by local refinement; the reported cutoff is moved an immaterial distance into the feasible interval, and the resulting value and constraints are checked numerically. These benchmarks recover the exact threshold-Grim frontier up to numerical root-finding tolerance. The public-forgiving benchmarks search a deterministic two-dimensional grid over the entry and return thresholds and then refine locally around the best feasible pair. Accordingly, the public-forgiving figures and output tables report the best feasible values found numerically, not theoretical maxima.

F(vi) Robustness and numerical caveats

As a time-window robustness check, I re-estimate the unchanged model using the final four available supergames in each session, or all available supergames when a session contains fewer than four.

Table 3: Final-four-supergames robustness

β	Usable noninitial choices	Mixture-estimated share (%)			
		AllD	Grim	Forgiving	Convergence
0	1,150	4.3	7.0	88.6	0
1	1,048	24.8	6.1	69.0	0
4	2,160	2.3	5.2	92.5	10
10	1,356	13.6	0.1	86.3	10

Estimated subject values and benchmarks: Grim

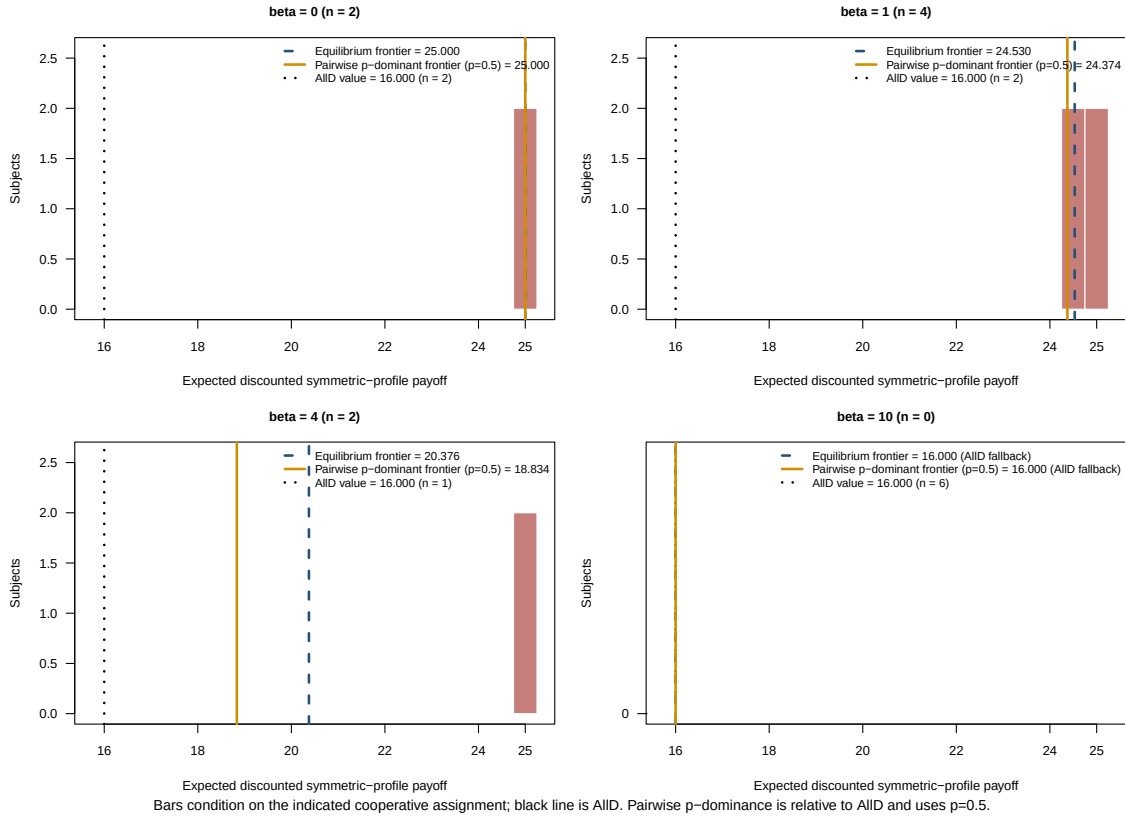


Figure 3: Final-four-supergames robustness for hard-assigned threshold-Grim subjects. Blue dashed lines show the exact equilibrium frontier, and gold solid lines impose, in addition, pairwise p -dominance at $p = 1/2$, relative to AllD. Black dotted lines mark the AllD value of 16 and report the number of hard-assigned AllD subjects.

Estimated subject values and benchmarks: Public forgiving

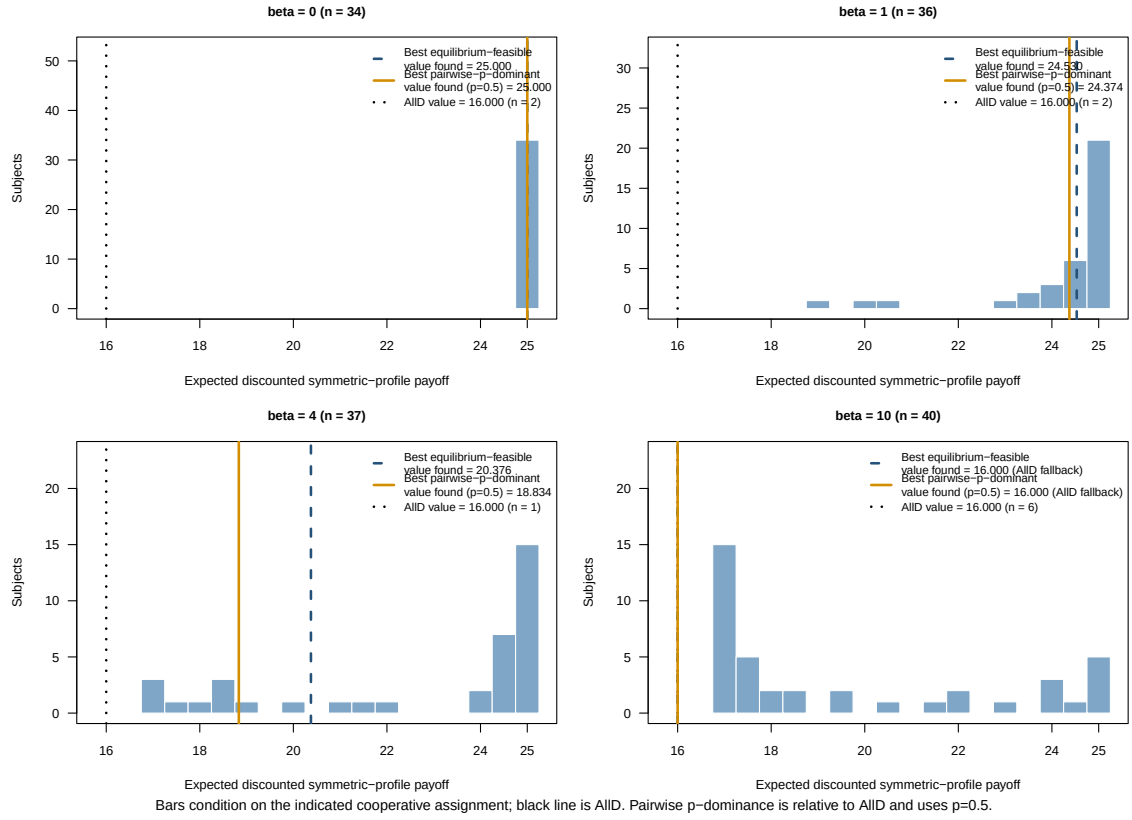


Figure 4: Final-four-supergames robustness for hard-assigned public-forgiving subjects. Blue dashed lines show the best equilibrium-feasible values; gold solid lines impose, in addition, pairwise p -dominance at $p = 1/2$, relative to AllID. Black dotted lines mark the AllID value of 16 and report the number of hard-assigned AllID subjects.

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